What I Built?

I completed all three assigned tasks as part of the internship assessment:

Task 1: Basic Image Manipulation

- Loaded an image using OpenCV.
- Performed grayscale conversion.
- Applied Gaussian Blur to smooth the image.
- Detected edges using the Canny Edge Detection algorithm.
- Saved all three processed images as separate output files.

Task 2: Mini Project — Color Detector

- Developed a Python script that opens a real-time webcam feed using OpenCV.
- Implemented a mouse click event using cv2.setMouseCallback() to capture the BGR color values at the clicked pixel.
- As a bonus, I also added a feature to display the closest color name using Euclidean distance from a predefined color list.
- Displayed a rectangle filled with the detected color and overlaid the color name and BGR values near the clicked region for a clean UI.

Challenges I Faced:

- Mapping BGR values to the closest named color required careful handling of NumPy arrays and efficient distance calculations.
- Making sure the rectangle and text overlay remained within the webcam frame without going off-screen was tricky.
- Setting up mouse callbacks in OpenCV was a new concept, and I had to understand how event handling works in real-time video processing.

What I Would Improve with More Time:

If I had more time, I would have focused on learning computer vision and OpenCV concepts in more depth, which would help me approach the project with better structure, cleaner code, and more optimized techniques. With a stronger foundation, I could explore advanced features like color space transformations, contour detection, or even add functionality like saving clicked color history and enhancing the user interface for a more polished result.

Refrences :-

https://chatgpt.com
Simple Color recognition with Opencv and Python
Detecting color with Python and OpenCV using HSV colorspace | Computer vision …
OpenCV tutorial for beginners | FULL COURSE in 3 hours with Python
https://docs.opencv.org/4.11.0/d6/d00/tutorial_py_root.html
https://docs.opencv.org/4.x/index.html
https://www.geeksforgeeks.org/multiple-color-detection-in-real-time-using-python-opencv
https://github.com/computervisioneng